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ORIGINAL ARTICLE

TESTING FOR CHANGE-POINTS IN LONG-RANGE DEPENDENT TIME SERIES BY MEANS OF A SELF-NORMALIZED WILCOXON TEST

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We propose a testing procedure based on the Wilcoxon two-sample test statistic in order to test for change-points in the mean of long-range dependent data. We show that the corresponding self-normalized test statistic converges in distribution to a non-degenerate limit under the hypothesis that no change occurred and that it diverges to infinity under the alternative of a change-point with constant height. Furthermore, we derive the asymptotic distribution of the self-normalized Wilcoxon test statistic under local alternatives, that is, under the assumption that the height of the level shift decreases as the sample size increases. Regarding the finite sample performance, simulation results confirm that the self-normalized Wilcoxon test yields a consistent discrimination between hypothesis and alternative and that its empirical size is already close to the significance level for moderate sample sizes.

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1. INTRODUCTION

We consider a data set generated by a stochastic process $(X_i)_{i\geq 1}$,

$$X_i = \mu_i + \varepsilon_i,$$

where $(\mu_i)_{i\geq 1}$ are unknown constants and where $(\varepsilon_i)_{i\geq 1}$ is a stationary, long-range dependent (LRD, in short) process with mean zero, finite variance and marginal distribution function F. In particular, we assume that

$$\varepsilon_i = G(\xi_i),\ i \geq 1, \tag{1}$$

where $(\xi_i)_{i\geq 1}$ is a stationary Gaussian process with mean 0, variance 1 and long-range dependence, that is, with autocovariance function ρ satisfying

$$\rho(k) \sim k^{-D} L(k),\ \text{as } k \to \infty, \tag{2}$$

where $0 < D < 1$ (referred to as long-range dependence parameter) and where L is a slowly varying function. Furthermore, we suppose that $G : \mathbb{R} \longrightarrow \mathbb{R}$ is a measurable function with $\mathrm{E}\,(G(\xi_i)) = 0$.

Provided that the previous assumptions hold for observations $X_1, \ldots, X_n$, we wish to test the hypothesis

$$H : \mu_1 = \cdots = \mu_n$$

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against the alternative

$$A : \mu_1 = \cdots = \mu_k \neq \mu_{k+1} = \cdots = \mu_n$$

for some $k \in \{1, \ldots, n-1\}$.

The change-point problem has been widely studied in the case of independent observations. A survey of parametric and non-parametric methods to detect change-points in i.i.d. data is given in Csörgő and Horváth (1997). Under the assumption of dependent linear processes, Horváth (1997) investigates the asymptotic properties of the Cusum change-point test. Giraitis *et al.* (1996) determine the asymptotic behaviour of Kolmogorov–Smirnov type change-point tests and change-point estimators for long-memory moving average processes. In this article, we consider a self-normalized rank test that serves to identify change-points in the mean of LRD data. So-called max-type rank tests are introduced by Antoch *et al.* (2008) for detecting changes in the distribution function of i.i.d. random variables. Wang (2008) studies change-point tests based on rank statistics considering infinite moving average long-memory time series. Dehling *et al.* (2013a) propose a change-point test based on the Wilcoxon two-sample rank test, which can be applied to detect changes in the mean of LRD processes that have a representation as nonlinear functionals of stationary Gaussian processes.

For the test problem (H, A), the location of the change-point is unknown under the alternative. In order to motivate our choice of a change-point test, we temporarily assume that the change-point location is known. That is, for a given $k \in \{1, \ldots, n-1\}$, we consider the alternative

$$A_k : \mu_1 = \cdots = \mu_k \neq \mu_{k+1} = \cdots = \mu_n.$$

Within this setting, the Wilcoxon two-sample rank test rejects the hypothesis of no change in the mean for large absolute values of the test statistic

$$W_{k,n} = \sum_{i=1}^{k} \sum_{j=k+1}^{n} \left(1_{\{X_i \leq X_j\}} - \frac{1}{2} \right).$$

The Wilcoxon change-point test for the test problem (H, A) is defined by reference to the test statistic $W_{k,n}$ (Dehling *et al.*, 2013a). It rejects the hypothesis for large values of

$$\max_{1 \leq k \leq n-1} |W_{k,n}| = \max_{1 \leq k \leq n-1} \left| \sum_{i=1}^{k} \sum_{j=k+1}^{n} \left(1_{\{X_i \leq X_j\}} - \frac{1}{2} \right) \right|.$$

With the objective of calculating the asymptotic distribution of the Wilcoxon test statistic under the null hypothesis, Dehling *et al.* (2013a) consider the stochastic process

$$W_n(\lambda) = \frac{1}{n d_n} \sum_{i=1}^{\lfloor n\lambda \rfloor} \sum_{j=\lfloor n\lambda \rfloor + 1}^{n} \left(1_{\{X_i \leq X_j\}} - \int_{\mathbb{R}} F(x) dF(x) \right), \ 0 \leq \lambda \leq 1, \tag{3}$$

where d_n denotes an appropriate normalization. Assuming that F is continuous, the asymptotic distribution of W_n can be derived from the empirical process invariance principle of Dehling and Taqqu (1989) as shown in Dehling *et al.* (2013a). It turns out that both, the limit of W_n and the normalization d_n, depend on the Hermite expansion

$$1_{\{G(\xi_i) \leq x\}} - F(x) = \sum_{q=1}^{\infty} \frac{J_q(x)}{q!} H_q(\xi_i),$$

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where H_q denotes the qth-order Hermite polynomial and where

$$J_q(x) = \mathrm{E}\left(H_q(\xi_i)1_{\{G(\xi_i)\le x\}}\right).$$

The scaling factor d_n is defined by

$$d_n^2 = \mathrm{Var}\left(\sum_{j=1}^{n} H_m(\xi_j)\right),$$

where m designates the Hermite rank of the class of functions $\{1_{\{G(\xi_1)\le x\}} - F(x),\ x \in \mathbb{R}\}$ defined by

$$m := \min\{q \ge 1 : J_q(x) \ne 0 \text{ for some } x \in \mathbb{R}\}.$$

Presuming that the previous conditions hold and the long-range dependence parameter D meets the condition $0 < D < \frac{1}{m}$, the process $W_n(\lambda)$, $0 \le \lambda \le 1$, defined by (3), converges in distribution to

$$\frac{1}{m!}(Z_m(\lambda) - \lambda Z_m(1)) \int_{\mathbb{R}} J_m(x)dF(x),\ 0 \le \lambda \le 1,$$

where $(Z_m(\lambda))_{\lambda\in[0,1]}$ is an mth-order Hermite process, which is self-similar with parameter $H = 1-\frac{mD}{2} \in \left(\frac{1}{2}, 1\right)$. If $m = 1$, the Hermite process Z_m equals a standard fractional Brownian motion process with Hurst parameter $H = 1 - \frac{D}{2}$. We refer to Taqqu (1979) for a general definition of the Hermite process Z_m. An application of the continuous mapping theorem to the process W_n yields the asymptotic distribution of the Wilcoxon change-point test. More precisely, it has been proved by Dehling *et al.* (2013a) that under the hypothesis of no change in the mean, the Wilcoxon test statistic

$$\frac{1}{nd_n} \max_{1\le k\le n-1} \left|\sum_{i=1}^{k} \sum_{j=k+1}^{n} \left(1_{\{X_i \le X_j\}} - \frac{1}{2}\right)\right|$$

converges in distribution to

$$\sup_{0\le\lambda\le 1} \left|\frac{1}{m!}(Z_m(\lambda) - \lambda Z_m(1))\right| \left|\int_{\mathbb{R}} J_m(x)dF(x)\right|.$$

Furthermore, Dehling *et al.* (2013b) investigate the asymptotic behaviour of the Wilcoxon change-point test under the alternative with the objective of determining the height of the level shift in such a way that the power of the test is non-trivial. For this purpose, they consider local alternatives defined by

$$A_{\tau,h_n} : \mu_i = \begin{cases} \mu & \text{for } i = 1,\dots,\lfloor n\tau\rfloor \\ \mu + h_n & \text{for } i = \lfloor n\tau\rfloor + 1,\dots,n, \end{cases}$$

where $0 < \tau < 1$ and where $h_n \sim c\frac{d_n}{n}$, so that under the sequence of local alternatives A_{τ,h_n}, the height of the level shift decreases if the sample size increases. Under the additional assumption that $G(\xi_i)$ has a continuous distribution function F with bounded density f, this guarantees that under the sequence of alternatives A_{τ,h_n}, the process

$$\frac{1}{nd_n} \sum_{i=1}^{\lfloor n\lambda\rfloor} \sum_{j=\lfloor n\lambda\rfloor+1}^{n} \left(1_{\{X_i \le X_j\}} - \frac{1}{2}\right),\ 0 \le \lambda \le 1,$$

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converges in distribution to the limit process

$$\frac{1}{m!}(Z_m(\lambda) - \lambda Z_m(1)) \int_{\mathbb{R}} J_m(x) dF(x) + c\delta_\tau(\lambda) \int_{\mathbb{R}} f^2(x) dx,\ 0 \le \lambda \le 1,$$

where $\delta_\tau : [0, 1] \longrightarrow \mathbb{R}$ is defined by

$$\delta_\tau(\lambda) = \begin{cases} \lambda(1-\tau) & \text{for } \lambda \le \tau \\ (1-\lambda)\tau & \text{for } \lambda \ge \tau \end{cases}.$$

By another application of the continuous mapping theorem, it then follows that the Wilcoxon change-point test statistic converges in distribution to a non-degenerate limit process under the sequence of local alternatives A_{τ,h_n} (Dehling *et al.*, 2013b).

An application of the Wilcoxon change-point test to a given data set presupposes the determination of the scaling factor d_n. Assuming that $0 < D < \frac{1}{m}$, we have $d_n^2 \sim c_m n^{2-mD} L^m(n)$, where $c_m = \frac{2m!}{(1-Dm)(2-Dm)}$ (Dehling *et al.*, 2013a). Consequently, the parameters D and m as well as the function L have to be known. Usually, this is not the case in statistical practice so that in general, the Wilcoxon change-point test as proposed in Dehling *et al.* (2013a) depends on an unknown normalization. While in many cases $m = 1$ and while there are methods to estimate D, estimating L seems to be hardly possible. For this reason, an approximation of d_n cannot be computed easily. As an alternative to normalizing the Wilcoxon test statistic by d_n, we propose a normalization that depends on the given realizations only and which is therefore referred to as self-normalization. Leaving aside the fact that the corresponding test statistic does not depend on unknown parameters, we will note that the expression $\int J_m(x) dF(x)$ (which is not known either) does not appear in its limit distribution. This is another advantage of the self-normalized Wilcoxon test as, in contrast, the limit distribution of the Wilcoxon change-point test statistic depends on this term.

The self-normalization approach we consider has originally been established in another context (Lobato, 2001). It has been extended to the change-point testing problem by Shao and Zhang (2010) in order to test for change-points in the mean of short-range dependent time series. These authors used the self-normalization method on the Kolmogorov–Smirnov test statistic, in doing so also taking the change-point alternative into account. Lobato as well as Shao and Zhang considered weakly dependent processes only. Following the approach in Shao and Zhang, an application to possibly LRD processes was introduced by Shao, who established a self-normalized version of a Cusum-like change-point test (Shao, 2011). As Cusum tests have the disadvantage of not being robust against possible outliers in the data, an extension of the self-normalization idea to the Wilcoxon test statistic leads to a change-point test that not only has the advantage of avoiding the choice of unknown parameters but also yields a robust alternative to Shao's self-normalized test.

The rest of the article is organized as follows. Section 2 introduces the self-normalized Wilcoxon test statistic and studies its limit behaviour under the hypothesis, fixed and certain local alternatives. Simulation results are presented in Section 3. Section 4 gives a short conclusion summarizing the main results of the article. Proofs of the theoretical results can be found in Appendix A.

2. MAIN RESULTS

Given observations $X_1, \ldots, X_n$, we consider the rank statistics defined by

$$R_i = \text{rank}(X_i) = \sum_{j=1}^{n} 1_{\{X_j \le X_i\}}$$

for $i = 1, \ldots, n$. An extension of the self-normalization approach to the Wilcoxon change-point test is based on an application of the Cusum change-point test in terms of the rank statistics R_i. Note that because of the identity

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$$\max_{1\le k\le n-1}\left|\sum_{i=1}^{k}R_i-\frac{k}{n}\sum_{i=1}^{n}R_i\right|=\max_{1\le k\le n-1}\left|\sum_{i=1}^{k}\sum_{j=k+1}^{n}\left(1_{\{X_i\le X_j\}}-\frac{1}{2}\right)\right|, \tag{4}$$

the Cusum test statistic of the ranks equals the Wilcoxon change-point test statistic. Instead of dividing the test statistic (which is the maximum taken among every possible outcome of the Wilcoxon two-sample rank test) by the unknown quantity nd_n, we consider a normalization that depends on the location of a potential change-point and that therefore is different for every possible outcome of the Wilcoxon two-sample rank test. We define

$$G_n(k)=\frac{\sum_{i=1}^{k}R_i-\frac{k}{n}\sum_{i=1}^{n}R_i}{\left\{\frac{1}{n}\sum_{t=1}^{k}S_t^2(1,k)+\frac{1}{n}\sum_{t=k+1}^{n}S_t^2(k+1,n)\right\}^{\frac{1}{2}}},$$

where

$$S_t(j,k)=\sum_{h=j}^{t}\left(R_h-\bar{R}_{j,k}\right),$$

$$\bar{R}_{j,k}=\frac{1}{k-j+1}\sum_{t=j}^{k}R_t.$$

The self-normalized Wilcoxon test rejects the hypothesis $H:\mu_1=\cdots=\mu_n$ for large values of the test statistic

$$T_n(\tau_1,\tau_2)=\sup_{k\in\{\lfloor n\tau_1\rfloor,\ldots,\lfloor n\tau_2\rfloor\}}|G_n(k)|,$$

where $0<\tau_1<\tau_2<1$.

Note that the proportion of the data that is included in the calculation of the supremum is restricted by τ_1 and τ_2. This is important as the choice of τ_1 and τ_2 influences the properties of the test. Structural breaks at the beginning or end of a sample are hard to detect since there is a lack of information concerning the behaviour of the time series before or after a potential break point. Hence, the interval $[\tau_1,\tau_2]$ must be small enough for the critical values not to get too large on the one hand, yet large enough to include potential break points on the other hand. A common choice is $\tau_1=1-\tau_2=0.15$ (Andrews, 1993).

Theorem 1 states the asymptotic distribution of the test statistic $T_n(\tau_1,\tau_2)$ under the hypothesis of no change in the mean.

Theorem 1. Let $(\xi_i)_{i\ge1}$ be a stationary, LRD Gaussian process with mean 0, variance 1 and LRD parameter D, and let $G:\mathbb{R}\longrightarrow\mathbb{R}$ be a measurable function. Suppose that $G(\xi_i)$ has a continuous distribution function F and that the stochastic process $(X_i)_{i\ge1}$ is defined by

$$X_i=\mu_i+G(\xi_i)$$

for unknown constants $(\mu_i)_{i\ge1}$. Moreover, assume that $0<D<\frac{1}{m}$, where m denotes the Hermite rank of the class of functions $1_{\{G(\xi_i)\le x\}}-F(x)$, $x\in\mathbb{R}$, and that $\int_{\mathbb{R}}J_m(x)dF(x)\neq0$. Then, under the hypothesis of no change in the mean, it follows that $T_n(\tau_1,\tau_2)\xrightarrow{\mathcal{D}}T(m,\tau_1,\tau_2)$, where

$$T(m,\tau_1,\tau_2)=\sup_{\lambda\in[\tau_1,\tau_2]}\frac{|Z_m(\lambda)-\lambda Z_m(1)|}{\left\{\int_0^{\lambda}V_m^2(r;0,\lambda)dr+\int_{\lambda}^{1}V_m^2(r;\lambda,1)dr\right\}^{\frac{1}{2}}}$$

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with

$$V_m(r; r_1, r_2) = Z_m(r) - Z_m(r_1) - \frac{r - r_1}{r_2 - r_1}\{Z_m(r_2) - Z_m(r_1)\}$$

for $r \in [r_1, r_2]$, $0 < r_1 < r_2 < 1$.

Remark 1. If $m = 1$, the process $Z_m(\lambda)$, $0 \le \lambda \le 1$, equals the standard fractional Brownian motion $B_H(\lambda)$, $0 \le \lambda \le 1$, with Hurst parameter $H = 1 - \frac{D}{2}$. In this case, the limit distribution of the self-normalized Wilcoxon test statistic resembles the limit distribution of the self-normalized test statistic introduced by Shao (2011). For one thing, this is due to the identity in equation (4) and because the self-normalized Wilcoxon test statistic results from an application of Shao's normalization approach to the rank statistics $R_1, \ldots, R_n$. For another thing, the limit distributions of the empirical process $d_n^{-1} \sum_{j=1}^{\lfloor n\lambda \rfloor} \left(1_{\{X_j \le x\}} - F(x)\right)$ and the partial sum process $d_n^{-1} \sum_{j=1}^{\lfloor n\lambda \rfloor} X_j$, which determine the asymptotic behaviour of the (self-normalized) Wilcoxon test statistic and (self-normalized) Cusum-like test statistics respectively, are equal up to a deterministic factor by reason of limit theorems in Dehling and Taqqu (1989) and Taqqu (1979).

As consistency under fixed alternatives is considered as a fundamental characteristic of appropriate hypothesis testing, we aim at proving Theorem 2, which implies that if there is a change-point in the mean of constant height, the empirical power of the self-normalized Wilcoxon test tends to 1. For this purpose, we suppose that under the alternative

$$X_i = \begin{cases} \mu + G(\xi_i), & i = 1, \ldots, k^*, \\ \mu + \Delta + G(\xi_i), & i = k^* + 1, \ldots, n, \end{cases} \tag{5}$$

where $k^* = \lfloor n\tau \rfloor$ and $\Delta \neq 0$ is fixed.

Theorem 2. Suppose that $(\xi_i)_{i \ge 1}$ is a stationary, LRD Gaussian process with mean 0, variance 1 and LRD parameter D. Moreover, let $G : \mathbb{R} \longrightarrow \mathbb{R}$ be a measurable function and assume that $G(\xi_i)$ has a continuous distribution function F. Given that the parameter D satisfies $0 < D < \frac{1}{m}$, where m denotes the Hermite rank of the class of functions $1_{\{G(\xi_1) \le x\}} - F(x)$, $x \in \mathbb{R}$, $T_n(\tau_1, \tau_2)$ diverges in probability to ∞ under fixed alternatives, that is, if $(X_i)_{i \ge 1}$ satisfies (5).

Furthermore, we wish to study the asymptotic behaviour of the self-normalized Wilcoxon change-point test under local alternatives defined by

$$A_{\tau, h_n}(n) : \mu_i = \begin{cases} \mu, & i = 1, \ldots, \lfloor n\tau \rfloor, \\ \mu + h_n, & i = \lfloor n\tau \rfloor + 1, \ldots, n, \end{cases}$$

where $0 < \tau < 1$ and $h_n \longrightarrow 0$.

Theorem 3 confirms that the self-normalized Wilcoxon test statistic converges to a non-degenerate limit under the sequence of local alternatives A_{τ, h_n}.

Theorem 3. Suppose that $(\xi_i)_{i \ge 1}$ is a stationary Gaussian process with mean 0, variance 1 and autocovariance function

$$\rho(k) \sim k^{-D} L(k),$$

where L is a slowly varying function and where $0 < D < \frac{1}{m}$. Moreover, let $G : \mathbb{R} \longrightarrow \mathbb{R}$ be a measurable function. We assume that $G(\xi_i)$ has a continuous distribution function F with bounded density f. Let m denote

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the Hermite rank of the class of functions $1_{\{G(\xi_1)\leq x\}} - F(x)$, $x \in \mathbb{R}$, and suppose that $\int_{\mathbb{R}} J_m(x)dF(x) \neq 0$. Then, under the sequence of alternatives A_{τ,h_n} with $h_n \sim c\frac{d_n}{n}$, it follows that $T_n(\tau_1,\tau_2)$ converges in distribution to

$$\tilde{T}_\tau(m,\tau_1,\tau_2) = \sup_{\lambda\in[\tau_1,\tau_2]} \frac{\left|\tilde{V}_{m,\tau}(\lambda;0,1)\right|}{\left\{\int_0^\lambda \tilde{V}^2_{m,\tau}(r;0,\lambda)dr + \int_\lambda^1 \tilde{V}^2_{m,\tau}(r;\lambda,1)dr\right\}^{\frac{1}{2}}}, \tag{6}$$

where

$$\tilde{V}_{m,\tau}(r;r_1,r_2) = \frac{1}{m!}\int_{\mathbb{R}} J_m(x)dF(x)V_m(r;r_1,r_2) + c\int_{\mathbb{R}} f^2(x)dx\left(\delta_\tau(r) - \delta_\tau(r_1) - \frac{r-r_1}{r_2-r_1}\{\delta_\tau(r_2)-\delta_\tau(r_1)\}\right)$$

with $V_m(r;r_1,r_2)$ as defined in Theorem 1.

Remark 2. The asymptotic behaviour of the self-normalized Wilcoxon test statistic is derived from Theorem 1.1 in Dehling *et al.* (2013a), that is, from the asymptotic behaviour of the Wilcoxon process W_n. The proof of this theorem is mainly based on the empirical process non-central limit theorem for subordinated Gaussian sequences in Dehling and Taqqu (1989). The sequential empirical process has also been studied for linear (or moving average) processes; a generalization of the limit theorem of Dehling and Taqqu is given by Theorem 1 in Giraitis and Surgailis (1999), for example. As a consequence, the asymptotic results, which have been proved for subordinated Gaussian processes in Dehling *et al.* (2013a) and in this article, are also applicable to the linear processes considered in Giraitis and Surgailis (1999).

3. SIMULATION STUDIES

We will now investigate the finite sample performance of the self-normalized Wilcoxon test statistic. For this purpose, we take $G(t) = t$ so that $(X_i)_{i\geq 1}$ is a Gaussian process. Since G is strictly increasing, the Hermite coefficient $J_1(x)$ is not equal to 0 for all $x \in \mathbb{R}$ (Dehling *et al.*, 2013a). Therefore, it holds that $m = 1$, where m denotes the Hermite rank of $1_{\{G(\xi_1)\leq x\}} - F(x), x \in \mathbb{R}$. As a result, $T_n(\tau_1,\tau_2)$ has approximately the same distribution as

$$\sup_{\lambda\in[\tau_1,\tau_2]} \frac{|B_H(\lambda) - \lambda B_H(1)|}{\left\{\int_0^\lambda V_H^2(r;0,\lambda)dr + \int_\lambda^1 V_H^2(r;\lambda,1)dr\right\}^{\frac{1}{2}}} \tag{7}$$

with

$$V_H(r;r_1,r_2) = B_H(r) - B_H(r_1) - \frac{r-r_1}{r_2-r_1}\{B_H(r_2) - B_H(r_1)\}, \tag{8}$$

where B_H is a fractional Brownian motion process with Hurst parameter $H = 1 - \frac{D}{2}$.

For the simulations, we will always choose $\tau_1 = 1 - \tau_2 = 0.15$. We set critical values for the asymptotic distribution of the self-normalized test statistic on the basis of 10,000 simulations of fractional Brownian motion time series for different Hurst parameters H and different levels of significance (Table I). Using the method `mvn` for the function `fbmSim` in the `fArma` package in `R`, fractional Brownian motion time series are generated from a numerical approximation of the stochastic integral.

For comparison with the self-normalized Wilcoxon test, we investigate the finite sample performance of the Wilcoxon test on the basis of the test statistic

$$\frac{1}{nd_n}\max_{k\in\{\lfloor n\tau_1\rfloor,\ldots,\lfloor n\tau_2\rfloor\}}\left|\sum_{i=1}^{k}\sum_{j=k+1}^{n}\left(1_{\{X_i\leq X_j\}} - \frac{1}{2}\right)\right|.$$

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Table I. Simulated critical values for the distribution of $T(1, \tau_1, \tau_2)$ when $[\tau_1, \tau_2] = [0.15, 0.85]$

	10%	5%	1%
$H = 0.501$	5.460569	6.429106	8.518842
$H = 0.6$	6.182835	7.276568	9.785915
$H = 0.7$	6.847260	8.190125	11.380584
$H = 0.8$	7.767277	9.495194	13.021080
$H = 0.9$	8.520039	10.333602	14.544094
$H = 0.999$	9.388174	11.339853	15.719645

The sample size is $\mathbf{1000}$; the number of replications is 10,000.

Table II. Upper quantiles of $\frac{1}{2\sqrt{\pi}} \sup_{\lambda \in [\tau_1, \tau_2]} |B_H(\lambda) - \lambda B_H(1)|$, where B_H is a standard fractional Brownian motion and $[\tau_1, \tau_2] = [0.15, 0.85]$

	10%	5%	1%
$H = 0.501$	0.3406971	0.3754267	0.4516422
$H = 0.6$	0.2732514	0.3063474	0.3761291
$H = 0.7$	0.2181069	0.2463102	0.3000083
$H = 0.8$	0.1669819	0.1875836	0.2335432
$H = 0.9$	0.1073921	0.1238605	0.1527903
$H = 0.999$	0.0100291	0.0115322	0.0143848

The sample size is $\mathbf{1000}$; the number of replications is 10,000.

Initially, we consider fractional Gaussian noise time series for the simulation study. In this case, the autocovariance function of $(X_i)_{i \geq 1}$ satisfies

$$\rho(k) \sim \left(1 - \frac{D}{2}\right)(1 - D)\, k^{-D}.$$

That is, the slowly varying function in (2) is given by $L(k) = \left(1 - \frac{D}{2}\right)(1 - D)$. Consequently, the normalization $n d_n$ is asymptotically equivalent to $n^{2-\frac{D}{2}}$. As $\int_{\mathbb{R}} J_1(x) dF(x) = -\frac{1}{2\sqrt{\pi}}$, the aforementioned test statistic converges in distribution to

$$\frac{1}{2\sqrt{\pi}} \sup_{\lambda \in [\tau_1, \tau_2]} |B_H(\lambda) - \lambda B_H(1)| \tag{9}$$

(Dehling *et al.*, 2013a). The critical values corresponding to the asymptotic distribution of the Wilcoxon test statistic in equation (9) are reported in Table II. The calculations of the relative frequency of false rejections under the hypothesis are based on 10,000 realizations of fractional Gaussian noise time series with varying length (see Table III for the Wilcoxon test and Table IV for the self-normalized Wilcoxon test).

The simulation results suggest that the self-normalized Wilcoxon test performs well under the hypothesis since empirical size and asymptotic significance level are already close for moderate sample sizes. In particular, it is notable that the size of the self-normalized Wilcoxon change-point test differs considerably from the size of the original Wilcoxon change-point test when $H = 0.9$, that means when we have very strong dependence. In that case, the convergence of the Wilcoxon change-point test statistic appears to be rather slow under the hypothesis, whereas the size of the self-normalized Wilcoxon change-point test is still close to the corresponding level of significance.

We consider fractional Gaussian noise time series with a level shift of height Δ after a proportion τ of the data in order to analyze the behaviour of the test statistics under the alternative. We have performed so for several choices of Δ and τ and for sample sizes $n = 100$ and $n = 500$ (see Table V for the Wilcoxon test and Table VI for the

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Table III. Level of the Wilcoxon change-point test for fractional Gaussian noise time series of length n with Hurst parameter H

n	$H = 0.6$	$H = 0.7$	$H = 0.8$	$H = 0.9$
10	0.016	0.023	0.095	0.320
50	0.047	0.044	0.065	0.173
100	0.050	0.047	0.060	0.146
500	0.057	0.054	0.057	0.096
1000	0.059	0.048	0.054	0.095

The level of significance is 5%. The calculations are based on 10,000 simulation runs.

Table IV. Level of the self-normalized Wilcoxon change-point test for fractional Gaussian noise time series of length n with Hurst parameter H

n	$H = 0.6$	$H = 0.7$	$H = 0.8$	$H = 0.9$
10	0.059	0.056	0.032	0.029
50	0.046	0.053	0.047	0.055
100	0.053	0.054	0.050	0.057
500	0.053	0.053	0.051	0.054
1000	0.055	0.054	0.052	0.057

The level of significance is 5%. The calculations are based on 10,000 simulation runs.

Table V. Empirical power of the Wilcoxon change-point test for fractional Gaussian noise of length $n = 100$ and $n = 500$ with Hurst parameter H and a level shift in the mean of height Δ after a proportion τ

		$\tau = 0.5$			$\tau = 0.25$		
	n	$\Delta = 0.5$	$\Delta = 1$	$\Delta = 2$	$\Delta = 0.5$	$\Delta = 1$	$\Delta = 2$
$H = 0.6$	100	0.348	0.916	1.000	0.204	0.690	1.000
	500	0.898	1.000	1.000	0.678	0.999	1.000
$H = 0.7$	100	0.239	0.696	0.999	0.125	0.401	0.968
	500	0.530	0.985	1.000	0.309	0.891	1.000
$H = 0.8$	100	0.177	0.526	0.984	0.106	0.265	0.839
	500	0.297	0.802	1.000	0.164	0.530	0.995
$H = 0.9$	100	0.287	0.628	0.983	0.198	0.403	0.889
	500	0.280	0.689	0.997	0.186	0.442	0.966

The level of significance is 5%. The calculations are based on 5000 simulation runs.

self-normalized Wilcoxon test). The simulations of the empirical power confirm that the rejection rate becomes higher when Δ increases. Comparing the empirical power for different Hurst parameters H, we note that the tests tend to have less power as H becomes large. This seems natural since when there is very strong dependence, that is, H is large, the variance of the series increases, so that it becomes harder to detect a level shift of a fixed height. In addition, change-points that are located in the middle of the sample are detected more often than change-points that are located close to the boundary of the testing region determined by $[\tau_1, \tau_2]$. Furthermore, the simulations show that an increasing sample size goes along with an increase of the empirical power. This result confirms that both change-point tests yield a consistent discrimination between hypothesis and alternative.

The calculations of the finite sample size and the empirical power of the standard Wilcoxon change-point test and the self-normalized Wilcoxon change-point test are based on simulations that were executed under the assumption that the Hurst parameter H is known. In practice, however, H has to be estimated from the data. Thus, we repeat the simulations, this time replacing H by the local Whittle estimator $\hat{H}$ proposed in Künsch (1987). Given

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Table VI. Empirical power of the self-normalized Wilcoxon change-point test for fractional Gaussian noise of length $n = 100$ and $n = 500$ with Hurst parameter H and a level shift in the mean of height Δ after a proportion τ

		$\tau = 0.5$			$\tau = 0.25$		
	n	$\Delta = 0.5$	$\Delta = 1$	$\Delta = 2$	$\Delta = 0.5$	$\Delta = 1$	$\Delta = 2$
$H = 0.6$	100	0.290	0.745	0.998	0.140	0.416	0.849
	500	0.712	0.996	1.000	0.385	0.865	0.998
$H = 0.7$	100	0.169	0.511	0.960	0.099	0.228	0.611
	500	0.366	0.860	1.000	0.200	0.523	0.904
$H = 0.8$	100	0.123	0.339	0.836	0.071	0.141	0.345
	500	0.180	0.541	0.969	0.092	0.234	0.605
$H = 0.9$	100	0.115	0.304	0.766	0.075	0.113	0.225
	500	0.145	0.387	0.881	0.079	0.149	0.350

The level of significance is 5%. The calculations are based on 5000 simulation runs.

observations $X_1, \ldots, X_n$, the estimation of the Hurst parameter is based on the values of the periodogram

$$I_X(\omega) = \frac{1}{2\pi n}\left|\sum_{j=1}^{n} X_j \mathrm{e}^{-ij\omega}\right|^2$$

at the Fourier frequencies

$$\omega_j = \frac{2\pi j}{n},\ j = 1, \ldots, m$$

with $m = m(n)$ denoting the spectral bandwidth, that is, the number of low frequencies used for the estimation. The local Whittle estimate is then defined by

$$\hat{H} = \operatorname{argmin}_{H \in [\kappa_1, \kappa_2]} R(H),$$

where

$$R(H) = \log\left(\frac{1}{m}\sum_{j=1}^{m} \omega_j^{2H-1} I_X(\omega_j)\right) - (2H-1)\frac{1}{m}\sum_{j=1}^{m} \log \omega_j$$

and with $0 < \kappa_1 < \kappa_2 < 1$ determining the domain of admissible estimates for H. In case that the bandwidth m satisfies $\frac{1}{m} + \frac{m}{n} \longrightarrow 0$ as $n \longrightarrow \infty$, that is, in case m diverges but diverges at a smaller rate than n, $\hat{H}$ is a consistent estimator for H (Robinson, 1995). The selection of the bandwidth has been addressed by several authors (e.g. Henry and Robinson 1996, Delgado and Robinson 1996 and Henry 2001). Still, for convenience, we always choose $m = \lfloor n^{2/3} \rfloor$ in this article. As we consider LRD time series characterized by an Hurst parameter H with $0.5 < H < 1$, we fix $\kappa_1 = 0.501$ and $\kappa_2 = 0.999$. In the implementation of the tests, critical values corresponding to $\hat{H}$ are therefore determined by a linear interpolation based on Table II for the Wilcoxon test and Table I for the self-normalized Wilcoxon test respectively. For each fractional Gaussian noise time series that underlies the simulated values of level and power, we calculate the local Whittle estimators $\hat{H}_1, \ldots, \hat{H}_n$ (with n denoting the number of repetitions for each simulation run). The average values of the biases, the sample standard deviations and the square roots of the sample mean squared error defined by

$$\text{Bias} = \bar{H} - H_0, \quad \hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(\hat{H}_i - \bar{H})^2}, \quad \sqrt{\text{MSE}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{H}_i - H_0)^2},$$

where $\bar{H} = \frac{1}{n}\sum_{i=1}^{n} \hat{H}_i$, are calculated under the hypothesis (Table VII) and under the alternative (Table VIII).

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Table VII. Estimation results for the local Whittle estimator with bandwidth $m = \lfloor n^{2/3} \rfloor$

n		$H = 0.6$	$H = 0.7$	$H = 0.8$	$H = 0.9$
10	Bias	0.089	0.030	−0.028	−0.089
	$\hat{\sigma}$	0.209	0.216	0.216	0.210
	$\sqrt{\text{MSE}}$	0.227	0.218	0.218	0.228
50	Bias	0.031	0.012	−0.005	−0.031
	$\hat{\sigma}$	0.138	0.159	0.161	0.146
	$\sqrt{\text{MSE}}$	0.142	0.160	0.161	0.149
100	Bias	0.018	0.006	0.003	−0.014
	$\hat{\sigma}$	0.110	0.129	0.132	0.117
	$\sqrt{\text{MSE}}$	0.111	0.129	0.132	0.118
500	Bias	0.001	−0.001	0.002	0.004
	$\hat{\sigma}$	0.066	0.072	0.074	0.068
	$\sqrt{\text{MSE}}$	0.066	0.072	0.074	0.068
1000	Bias	−0.003	0.001	0.003	0.006
	$\hat{\sigma}$	0.053	0.056	0.056	0.054
	$\sqrt{\text{MSE}}$	0.053	0.056	0.056	0.055

The calculations are based on 10,000 simulation runs of fractional Gaussian noise time series of length n.

Table VIII. Estimation results for the local Whittle estimator with bandwidth $m = \lfloor n^{2/3} \rfloor$

		$\tau = 0.5$				$\tau = 0.25$			
Δ		$H = 0.6$	$H = 0.7$	$H = 0.8$	$H = 0.9$	$H = 0.6$	$H = 0.7$	$H = 0.8$	$H = 0.9$
0.5	Bias	0.106	0.059	0.037	0.024	0.084	0.048	0.029	0.021
	$\hat{\sigma}$	0.062	0.070	0.073	0.065	0.066	0.071	0.074	0.065
	$\sqrt{\text{MSE}}$	0.123	0.092	0.082	0.070	0.106	0.086	0.079	0.069
1	Bias	0.233	0.156	0.102	0.063	0.205	0.136	0.088	0.058
	$\hat{\sigma}$	0.046	0.057	0.065	0.048	0.052	0.062	0.067	0.051
	$\sqrt{\text{MSE}}$	0.237	0.166	0.121	0.080	0.212	0.149	0.111	0.077
2	Bias	0.382	0.284	0.191	0.098	0.365	0.270	0.184	0.096
	$\hat{\sigma}$	0.023	0.023	0.020	0.007	0.032	0.033	0.028	0.012
	$\sqrt{\text{MSE}}$	0.383	0.285	0.192	0.098	0.366	0.272	0.186	0.097

The calculations are based on 5000 simulation runs of fractional Gaussian noise time series of length 500 with a level shift in the mean of height Δ after a proportion τ.

Regardless of small samples, we can see from Table VII that, on average, the local Whittle estimator tends to slightly overestimate the true value of H if fractional Gaussian noise time series are considered. Therefore, the corresponding critical values in Table I will usually be bigger than the critical values that correspond to H, so that we expect the self-normalized Wilcoxon test to reject the hypothesis less often. Comparing the rejection rates of the self-normalized Wilcoxon test under the hypothesis for an estimated H (Table IX) with the previous simulation results (Table IV), we note that although the test tends to reject the hypothesis less often when H is estimated, the deviations of the empirical probabilities are negligible. For comparison, we also calculate the empirical size of the Wilcoxon change-point test for an estimated Hurst parameter (Table X). As regards the critical values corresponding to $\hat{H}$, we expect the Wilcoxon test to reject the hypothesis more often since the critical values that result from an estimation $\hat{H}$ of H, which is bigger than H, are smaller than the critical values that correspond to the true value of H. Nevertheless, our simulation results show that (if we disregard small sample sizes) the Wilcoxon change-point test rejects the hypothesis less often for Hurst parameters H less than 0.9 (Table X). This is because instead of normalizing the test statistic by $n d_n$, which is $n^{2-\frac{D}{2}}$ for fractional Gaussian noise time series, we divide the Wilcoxon test statistic by $n^{2-\frac{\hat{D}}{2}}$, where $\hat{D} = 2 - 2\hat{H}$, so that the test statistic is scaled down too fast. However, if $H = 0.9$, estimating H by the local Whittle estimator leads to a larger number of false rejections

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Table IX. Level of the self-normalized Wilcoxon change-point test for fractional Gaussian noise time series of length n with Hurst parameter H, estimated by the local Whittle estimator with bandwidth $m = \lfloor n^{2/3} \rfloor$

n	$H = 0.6$	$H = 0.7$	$H = 0.8$	$H = 0.9$
10	0.022	0.027	0.028	0.028
50	0.034	0.034	0.047	0.049
100	0.041	0.050	0.050	0.056
500	0.051	0.051	0.048	0.054
1000	0.052	0.051	0.051	0.052

The level of significance is 5%. The calculations are based on 10,000 simulation runs.

Table X. Level of the Wilcoxon change-point test for fractional Gaussian noise time series of length n with Hurst parameter H, estimated by the local Whittle estimator with bandwidth $m = \lfloor n^{2/3} \rfloor$

n	$H = 0.6$	$H = 0.7$	$H = 0.8$	$H = 0.9$
10	0.266	0.345	0.433	0.518
50	0.039	0.125	0.280	0.482
100	0.021	0.061	0.203	0.450
500	0.036	0.035	0.051	0.317
1000	0.046	0.036	0.039	0.246

The level of significance is 5%. The calculations are based on 10,000 simulation runs.

under the hypothesis. The mostly negative biases indicate that, in this case, the opposite effect holds: as a result of a normalization depending on $\hat{H}$, which is smaller than the original normalization, the test statistic is scaled down too slowly, and therefore, the test falsely rejects the hypothesis. For small samples, the biases as well as the values of the sample standard deviation $\hat{\sigma}$ reported in Table VII verify that the deviations of the local Whittle estimator from the true value of H tend to be exceptionally large. This may explain the relatively high rejection rate of the Wilcoxon change-point test in these cases. Furthermore, the impact of smaller critical values easily outreaches the effect of a stronger normalization in small samples and may therefore lead to spurious rejections of the hypothesis.

The presence of change-points in time series causes the local Whittle estimator to overestimate H (Table VIII). In particular, one notes that the higher the level shift, the larger the estimated value and the larger the deviation from the nominal value of H. As a result, for the Wilcoxon change-point test, the critical values corresponding to $\hat{H}$ will be smaller than the critical values corresponding to the true value of H. Therefore, we may expect the Wilcoxon test to detect change-points more often. On the other hand, replacing H by the local Whittle estimator has the effect of scaling down the test statistic too heavily, so that we may also expect the test to reject the hypothesis less often. Actually, our simulation results confirm that replacing D by $\hat{D}$ in the normalization has the greater impact on the empirical power of the Wilcoxon change-point test since the rejection rate under the alternative is in most cases smaller if H is estimated (Table XI). Again, the case $H = 0.9$ is an exception since an estimation of H goes along with an increase in power of the Wilcoxon test. Concerning the self-normalized Wilcoxon test, we observe that estimating H entails a power loss (Table XII). This is because the overestimation of H under the alternative yields critical values that are bigger than the critical values corresponding to the true value of H.

The previous considerations already emphasized the benefits that go along with an application of the self-normalized Wilcoxon change-point test compared with the usage of the Wilcoxon change-point test proposed by Dehling *et al.* (2013a). In the following, we will also note the advantages over the self-normalized change-point test introduced by Shao (2011).

It is well known that the Wilcoxon change-point test is more robust against outliers in data sets than the Cusum-like difference-of-means test. That is, one expects the Wilcoxon test to outperform the difference-of-means test if heavy-tailed time series are considered. We consider Pareto-distributed data, so as to confirm that this observation also holds for the self-normalized versions of the difference-of-means test and the Wilcoxon change-point test.

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Table XI. Empirical power of the Wilcoxon change-point test for fractional Gaussian noise of length $n = 500$ with Hurst parameter H, estimated by the local Whittle estimator with bandwidth $m = \lfloor n^{2/3} \rfloor$, and a level shift in the mean of height Δ after a proportion τ

	$\tau = 0.5$			$\tau = 0.25$		
	$\Delta = 0.5$	$\Delta = 1$	$\Delta = 2$	$\Delta = 0.5$	$\Delta = 1$	$\Delta = 2$
$H = 0.6$	0.374	0.900	1.000	0.142	0.250	1.000
$H = 0.7$	0.224	0.781	1.000	0.085	0.259	0.996
$H = 0.8$	0.229	0.722	0.999	0.121	0.411	0.986
$H = 0.9$	0.517	0.846	0.999	0.451	0.753	0.996

The level of significance is 5%. The calculations are based on 5000 simulation runs.

Table XII. Empirical power of the self-normalized Wilcoxon change-point test for fractional Gaussian noise of length $n = 500$ with Hurst parameter H, estimated by the local Whittle estimator with bandwidth $m = \lfloor n^{2/3} \rfloor$, and a level shift in the mean of height Δ after a proportion τ

	$\tau = 0.5$			$\tau = 0.25$		
	$\Delta = 0.5$	$\Delta = 1$	$\Delta = 2$	$\Delta = 0.5$	$\Delta = 1$	$\Delta = 2$
$H = 0.6$	0.577	0.958	1.000	0.303	0.687	0.933
$H = 0.7$	0.302	0.730	0.992	0.133	0.359	0.742
$H = 0.8$	0.163	0.462	0.924	0.085	0.198	0.461
$H = 0.9$	0.128	0.350	0.834	0.068	0.120	0.289

The level of significance is 5%. The calculations are based on 5000 simulation runs.

Table XIII. Level of the self-normalized Cusum change-point test for standardized Pareto(3,1)-transformed fractional Gaussian noise time series of length n with Hurst parameter H

n	$H = 0.6$	$H = 0.7$	$H = 0.8$	$H = 0.9$
10	0.144	0.116	0.095	0.072
50	0.143	0.113	0.090	0.069
100	0.151	0.121	0.097	0.070
500	0.139	0.106	0.083	0.060
1000	0.143	0.108	0.081	0.057

The level of significance is 5%. The calculations are based on 10,000 simulation runs.

In order to obtain standardized Pareto-distributed data, which has a representation as a functional of a Gaussian process, we consider the transformation

$$G(t) = \left(\frac{\beta k^2}{(\beta - 1)^2 (\beta - 2)} \right)^{-\frac{1}{2}} \left(k(\Phi(t))^{-\frac{1}{\beta}} - \frac{\beta k}{\beta - 1} \right)$$

with parameters $k, \beta > 0$ and with Φ denoting the standard normal distribution function. Since G is a strictly decreasing function, it follows by Theorem 2 in Dehling *et al.* (2013a) that the Hermite rank of $1_{\{G(\xi_1) \leq x\}} - F(x), x \in \mathbb{R}$, is $m = 1$. Hence, the limit distribution of the self-normalized Wilcoxon test statistic corresponds to the expression in equation (7), while the limit distribution of Shao's test statistic is given in Theorem 1 in Shao (2011).

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The calculations of the relative frequency of false rejections under the hypothesis are based on realizations of Pareto-distributed time series with parameters $\beta = 3$ and $k = 1$ and sample sizes ranging from $n = 10$ to $n = 1000$ (see Table XIII for the level of the self-normalized Cusum test and Table XIV for the level of the self-normalized Wilcoxon test). As regards the empirical size of the self-normalized test statistics, it should be noted that the self-normalized Wilcoxon test performs well already for moderate sample sizes (Table XIV), while the self-normalized Cusum-like test statistic even fails to get close to the asymptotic level of 5% for a sample size of $n = 1000$ (Table XIII). Furthermore, a comparison of Table XV and Table XVI shows that the self-normalized Wilcoxon test has larger power than Shao's self-normalized change-point test for most of the considered level shifts and break points. Even though Shao's self-normalized change-point test tends to have larger power than the self-normalized Wilcoxon change-point test for high-level shifts that are located after a proportion $\tau = 0.25$ of

Table XIV. Level of the self-normalized Wilcoxon change-point test for standardized Pareto(3,1)-transformed fractional Gaussian noise time series of length n with Hurst parameter H

n	$H = 0.6$	$H = 0.7$	$H = 0.8$	$H = 0.9$
10	0.059	0.056	0.032	0.029
50	0.046	0.053	0.047	0.055
100	0.053	0.054	0.050	0.057
500	0.053	0.056	0.048	0.056
1000	0.056	0.055	0.053	0.054

The level of significance is 5%. The calculations are based on 10,000 simulation runs.

Table XV. Empirical power of the self-normalized Cusum change-point test for standardized Pareto(3,1)-transformed fractional Gaussian noise time series of length $n = 500$ with Hurst parameter H and a level shift in the mean of height Δ after a proportion τ

	$\tau = 0.5$			$\tau = 0.25$		
	$\Delta = 0.5$	$\Delta = 1$	$\Delta = 2$	$\Delta = 0.5$	$\Delta = 1$	$\Delta = 2$
$H = 0.6$	0.676	0.978	0.999	0.579	0.947	0.997
$H = 0.7$	0.420	0.877	0.995	0.309	0.788	0.987
$H = 0.8$	0.284	0.675	0.960	0.189	0.559	0.903
$H = 0.9$	0.295	0.604	0.878	0.209	0.499	0.808

The level of significance is 5%. The calculations are based on 5000 simulation runs.

Table XVI. Empirical power of the self-normalized Wilcoxon change-point test for standardized Pareto(3,1)-transformed fractional Gaussian noise time series of length $n = 500$ with Hurst parameter H and a level shift in the mean of height Δ after a proportion τ

	$\tau = 0.5$			$\tau = 0.25$		
	$\Delta = 0.5$	$\Delta = 1$	$\Delta = 2$	$\Delta = 0.5$	$\Delta = 1$	$\Delta = 2$
$H = 0.6$	0.999	1.000	1.000	0.951	0.998	1.000
$H = 0.7$	0.951	0.999	1.000	0.696	0.912	0.964
$H = 0.8$	0.702	0.932	0.994	0.356	0.570	0.728
$H = 0.9$	0.534	0.809	0.945	0.209	0.346	0.432

The level of significance is 5%. The calculations are based on 5000 simulation runs.

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the considered time series (see Table XV for the power of the self-normalized Cusum test and Table XVI for the power of the self-normalized Wilcoxon test), this must not be overrated since the realized level of Shao's test is considerably larger than the level of significance, while the self-normalized Wilcoxon change-point test has an empirical size of approximately 5%.

So far, the simulation results are based on time series with an autocovariance function characterized by a slowly varying function L that asymptotically behaves like a constant. Since estimating L does not seem to be an easy task, the fact that an application of self-normalized change-point tests does not require knowledge or estimation of L can be considered as a major advantage over tests with non-random normalization. For this reason, it may be interesting to consider the performance of the self-normalized change-point test in the presence of a slowly varying function L that is not constant. As an example of such time series, we take Gaussian observations with autocovariance function $\rho(k) = k^{-D}\ln(k+1)$, $k \geq 1$. In `R`, these are generated by the function `rmvnorm` in the `mvtnorm` package. In order to be able to apply the change-point tests, we assume that the given data are fractional Gaussian noise. Therefore, we choose $nd_n = n^{2-\frac{D}{2}}$ as normalization for the Wilcoxon test statistic, and we take the 5% quantiles of the limits in (9) and (7) as critical values. Note that for the classical Wilcoxon test, the limit distribution in (9) is the right choice, whereas the normalization of the test statistic is misspecified. Since the self-normalized test statistic can be computed from the given data only, there is no need to specify the normalization. That is, in this case, there is no misspecification at all. As the purpose of the simulations is to investigate the effect of the slowly varying function L on the performance of the change-point tests, estimation of the Hurst parameter is neglected. In particular, approximating H by the local Whittle estimator presupposes that the spectral density f of the considered time series satisfies $f(\lambda) \sim C\lambda^{D-1}$, $\lambda \to 0$, where C is a positive constant. Because of Corollary 1.1.15 in Pipiras and Taqqu (2011), this assumption does not hold for the considered time series. However, it is shown in Robinson (2014), for example, that feasible estimation of H is possible even if L is not constant. The simulation results presented in Tables XVII and XVIII show that for Gaussian observations with autocovariance function $\rho(k) = k^{-D}\ln(k+1)$, the performance of the change-point tests is not nearly as good as

Table XVII. Level of the Wilcoxon change-point test for long-range dependent Gaussian time series of length n with Hurst parameter H and with autocovariance function $\rho(k) = k^{2H-2}\ln(k+1)$, $k \geq 1$

n	$H = 0.6$	$H = 0.7$	$H = 0.8$	$H = 0.9$
10	0.054	0.021	0.070	0.427
50	0.520	0.346	0.004	0.319
100	0.674	0.517	0.141	0.241
500	0.900	0.783	0.533	0.020
1000	0.948	0.845	0.646	0.000

The level of significance is 5%. The calculations are based on 10,000 simulation runs.

Table XVIII. Level of the self-normalized Wilcoxon change-point test for long-range dependent Gaussian time series of length n with Hurst parameter H and with autocovariance function $\rho(k) = k^{2H-2}\ln(k+1)$, $k \geq 1$

n	$H = 0.6$	$H = 0.7$	$H = 0.8$	$H = 0.9$
10	0.138	0.047	0.041	0.048
50	0.205	0.229	0.012	0.076
100	0.193	0.203	0.137	0.055
500	0.147	0.153	0.163	0.009
1000	0.122	0.121	0.128	0.000

The level of significance is 5%. The calculations are based on 10,000 simulation runs.

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for fractional Gaussian noise time series. The size of both testing procedures differs considerably from the nominal level. Considering the finite sample performance of the classical Wilcoxon test, it is notable that a growing sample size goes along with an increase of the rejection rate under the hypothesis. This observation seems to contradict the expected behaviour of change-point tests, namely, that the empirical size converges to the level of significance. Apart from that, the rejection rate of the self-normalized test tends to approach the asymptotic level as the number of observations grows. In this regard and because, in general, the empirical size of the self-normalized testing procedure is closer to the asymptotic level of 5%, we conclude that the self-normalized Wilcoxon change-point test can be considered as being more robust to the slowly varying function L.

4. CONCLUSIONS

In this article, we investigate the properties of a change-point test, which results from an application of Shao's self-normalization approach to the Wilcoxon change-point test proposed by Dehling *et al.* (2013a). We derive the asymptotic distribution of the corresponding test statistic under the hypothesis and under local alternatives and prove consistency under fixed alternatives. The self-normalized test statistic is characterized by a normalization, which can be calculated on the basis of the observed data, while its limit distribution depends on the Hurst parameter only. In particular, the self-normalized testing procedure does not depend on a specification of the slowly varying function L that characterizes the autocovariance function of LRD time series. Our simulation results confirm that the self-normalized Wilcoxon test can be considered as being robust to the slowly varying function L, whereas the choice of L may severely affect the performance of the standard Wilcoxon change-point test.

Moreover, the simulations show that the self-normalized Wilcoxon change-point test reaches its asymptotic level for finite sample sizes of moderate lengths if LRD time series with an Hurst parameter varying from $H = 0.6$ to $H = 0.9$ are considered. In contrast, the Wilcoxon change-point test falsely rejects the hypothesis too often in case of fractional Gaussian noise time series with Hurst parameter $H = 0.9$. Concerning the empirical power, one has to note that the classical Wilcoxon change-point test tends to detect changes in the mean more often than the self-normalized Wilcoxon test. Nevertheless, the empirical power of the self-normalized Wilcoxon change-point test is still competitive. In the context of change-point tests, the so-called 'better size but less power' phenomenon for self-normalized test statistics has also been observed by Shao (2011) as well as Shao and Zhang (2010). Moreover, it is consistent with observations made by Lobato (2001) and Sun *et al.* (2008).

Furthermore, our simulation results confirm that an estimation of the Hurst parameter leads to an increase of the empirical size and a decrease of the empirical power for both tests, the standard Wilcoxon change-point test and the self-normalized Wilcoxon test. However, it should also be noted that the impact of an estimated Hurst parameter on the self-normalized Wilcoxon change-point test is rather small, whereas the finite sample performance of the standard Wilcoxon change-point test is affected by different effects that result from an estimation of H.

As regards heavy-tailed time series, our simulation results show that the self-normalized Wilcoxon change-point test is more robust against outliers in the data since it detects change-points more often than the self-normalized difference-of-means test proposed by Shao (2011) and rejects the hypothesis less often if there is no change-point in the data. In view of its finite sample performance, the self-normalized Wilcoxon change-point test should be considered as an alternative to existing change-point tests as it is able to compete with change-point tests like the classical Wilcoxon change-point test and the self-normalized change-point test proposed by Shao (2011).

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APPENDIX A

Proofs.

In order to simplify notation, we write

$$J(x) = \frac{1}{m!} J_m(x),$$
$$Z(\lambda) = Z_m(\lambda).$$

Proof of Theorem 1. The essential step in the proof of Theorem 1 is to find a representation for the test statistic $T_n(\tau_1, \tau_2)$ as a functional of the Wilcoxon process

$$W_n(\lambda) = \frac{1}{n d_n} \sum_{i=1}^{\lfloor n\lambda \rfloor} \sum_{j=\lfloor n\lambda \rfloor + 1}^{n} \left(1_{\{X_i \le X_j\}} - \frac{1}{2} \right), \ 0 \le \lambda \le 1.$$

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For this purpose, rewrite

$$G_n(k) = \frac{\frac{1}{nd_n}\sum_{i=1}^{k}\sum_{j=k+1}^{n}\left(1_{\{X_i \le X_j\}} - \frac{1}{2}\right)}{\frac{1}{nd_n}\left\{\frac{1}{n}\sum_{t=1}^{k} S_t^2(1,k) + \frac{1}{n}\sum_{t=k+1}^{n} S_t^2(k+1,n)\right\}^{\frac{1}{2}}}.$$

As the numerator of $G_n(k)$ equals $W_n(\lambda)$ if $k = \lfloor n\lambda \rfloor$, it remains to show that the denominator of $G_n(k)$ can be represented as a functional of W_n. Bearing in mind that the identity

$$\sum_{i=1}^{l}\sum_{j=1}^{l} 1_{\{X_i \le X_j\}} = \frac{l(l+1)}{2} \tag{A1}$$

holds almost surely, elementary calculations show that the properly normalized denominator of $G_n(k)$, with $k = \lfloor n\lambda \rfloor$, can (almost surely) be represented as follows:

$$\begin{aligned}
&\frac{1}{nd_n}\left\{\frac{1}{n}\sum_{t=1}^{k} S_t^2(1,k) + \frac{1}{n}\sum_{t=k+1}^{n} S_t^2(k+1,n)\right\}^{\frac{1}{2}} \\
&\quad = \left\{\int_0^{\lambda}\left(W_n(r) - \frac{\lfloor nr \rfloor}{\lfloor n\lambda \rfloor}W_n(\lambda)\right)^2 dr + \int_{\lambda}^{1}\left(W_n(r) - \frac{n - \lfloor nr \rfloor}{n - \lfloor n\lambda \rfloor}W_n(\lambda)\right)^2 dr\right\}^{\frac{1}{2}}.
\end{aligned}$$

All in all, this yields

$$T_n(\tau_1,\tau_2) = \sup_{\lambda \in [\tau_1,\tau_2]} \frac{|W_n(\lambda)|}{\left\{\int_0^{\lambda}\left(W_n(r) - \frac{\lfloor nr \rfloor}{\lfloor n\lambda \rfloor}W_n(\lambda)\right)^2 dr + \int_{\lambda}^{1}\left(W_n(r) - \frac{n-\lfloor nr \rfloor}{n-\lfloor n\lambda \rfloor}W_n(\lambda)\right)^2 dr\right\}^{\frac{1}{2}}}.$$

Consider the function $G : D[0,1] \longrightarrow \mathbb{R}$, which is defined by

$$G(f) = \sup_{\lambda \in [\tau_1,\tau_2]} |G_\lambda(f)|$$

with

$$G_\lambda(f) = \frac{f(\lambda)}{\left\{\int_0^{\lambda}\left(f(r) - \frac{r}{\lambda}f(\lambda)\right)^2 dr + \int_{\lambda}^{1}\left(f(r) - \frac{1-r}{1-\lambda}f(\lambda)\right)^2 dr\right\}^{\frac{1}{2}}},$$

provided that the function $H : D[0,1] \longrightarrow \mathbb{R}$ defined by

$$H(f) = \inf_{\lambda \in [\tau_1,\tau_2]}\left\{\int_0^{\lambda}\left(f(r) - \frac{r}{\lambda}f(\lambda)\right)^2 dr + \int_{\lambda}^{1}\left(f(r) - \frac{1-r}{1-\lambda}f(\lambda)\right)^2 dr\right\}^{\frac{1}{2}}$$

does not equal 0 in f. Given that $f \in H^{-1}(\{0\})$, we set $G_\lambda(f) = -1$. Since $\frac{\lfloor n\lambda \rfloor}{n} = \lambda + o(1)$, the limit of $T_n(\tau_1,\tau_2)$ is equal to the limit of $G(W_n(\cdot))$.

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Note that because of Theorem 1.1 in Dehling *et al.* (2013a), $W_n(\lambda)$, $0 \leq \lambda \leq 1$, converges in distribution towards the process

$$W^{(m)}(\lambda) = (Z(\lambda) - \lambda Z(1)) \int_{\mathbb{R}} J(x) dF(x),\ 0 \leq \lambda \leq 1.$$

In order to determine the asymptotic distribution of $T_n(\tau_1, \tau_2)$, we intend to apply the continuous mapping theorem. Thus, we have to prove the following assertions:

(1) The function G is measurable with respect to the uniform metric on $D\,[0,1]$.
(2) We have $P(W^{(m)} \in D_G) = 0$, where D_G denotes the set of discontinuities of G.

In order to show that G is measurable, we consider the restrictions of G to $D\,[0,1] \setminus H^{-1}(\{0\})$ and $H^{-1}(\{0\})$ respectively. Both restrictions are continuous with respect to the uniform metric. In particular, both restrictions are Borel measurable. Since the restricted domains are Borel measurable subsets of $D\,[0,1]$, the measurability of the restrictions implies the measurability of G. It remains to show that $P(W^{(m)} \in D_G) = 0$. As G is continuous at every $h \in D\,[0,1] \setminus H^{-1}(\{0\})$, D_G is a subset of $H^{-1}(\{0\})$. Consequently, it suffices to show that $P(W^{(m)} \in H^{-1}(\{0\})) = 0$. It can easily be seen that all continuous functions that lie in $H^{-1}(\{0\})$ are differentiable almost everywhere. Since the sample paths of $W^{(m)}$ are almost surely continuous but nowhere differentiable with probability 1 (Mikosch, 1998), it therefore follows that $P(W^{(m)} \in H^{-1}(\{0\})) = 0$.

Having verified the preconditions of the continuous mapping theorem, we are now able to conclude that the test statistic $T_n(\tau_1, \tau_2)$ converges in distribution to

$$T(m, \tau_1, \tau_2) = \sup_{\lambda \in [\tau_1, \tau_2]} \frac{\left|W^{(m)}(\lambda)\right|}{\left\{\int_0^\lambda \left(W^{(m)}(r) - \frac{r}{\lambda} W^{(m)}(\lambda)\right)^2 dr + \int_\lambda^1 \left(W^{(m)}(r) - \frac{1-r}{1-\lambda} W^{(m)}(\lambda)\right)^2 dr\right\}^{\frac{1}{2}}}.$$

It follows by elementary calculations that the limit process $T(m, \tau_1, \tau_2)$ equals

$$\sup_{\lambda \in [\tau_1, \tau_2]} \frac{|Z(\lambda) - \lambda Z(1)|}{\left\{\int_0^\lambda (V_m(r; 0, \lambda))^2\, dr + \int_\lambda^1 (V_m(r; \lambda, 1))^2\, dr\right\}^{\frac{1}{2}}}.$$

Thus, we have established Theorem 1. □

In the proof of Theorem 2, we make use of preliminary results stated in Lemmas 1 and 2. The line of argument that verifies the lemmas is a modification of the proof that establishes Theorem 3.1 in Dehling *et al.* (2013b).

Lemma 1. Suppose that $(\xi_i)_{i \geq 1}$ is a stationary, long-range dependent Gaussian process with mean 0, variance 1 and long-range dependence parameter $0 < D < \frac{1}{m}$, where m denotes the Hermite rank of the class of functions $1_{\{G(\xi_1) \leq x\}} - F(x)$, $x \in \mathbb{R}$. Moreover, assume that $(G(\xi_i))_{i \geq 1}$ has a continuous distribution function F and that $G : \mathbb{R} \longrightarrow \mathbb{R}$ is a measurable function. Then, if $\Delta \in \mathbb{R}$,

$$\frac{1}{n^2} \sum_{i=1}^{\lfloor n\lambda \rfloor} \sum_{j=\lfloor n\tau \rfloor + 1}^{n} 1_{\{G(\xi_i) \leq G(\xi_j) + \Delta\}} \xrightarrow{P} \lambda(1-\tau) \int_{\mathbb{R}} F(x+\Delta) dF(x),$$
$$\frac{1}{n^2} \sum_{i=1}^{\lfloor n\tau \rfloor} \sum_{j=\lfloor n\lambda \rfloor + 1}^{n} 1_{\{G(\xi_i) \leq G(\xi_j) + \Delta\}} \xrightarrow{P} \tau(1-\lambda) \int_{\mathbb{R}} F(x+\Delta) dF(x),$$

for fixed τ, uniformly in $\lambda \leq \tau$ and $\lambda \geq \tau$ respectively.

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Proof of Lemma 1. We give a proof for the first assertion only as the convergence of the second term follows by an analogous argumentation.

Let F_k and $F_{k+1,n}$ denote the empirical distribution functions of the first k and last $n-k$ realizations of $G(\xi_1), \dots, G(\xi_n)$, that is,

$$F_k(x) = \frac{1}{k} \sum_{i=1}^{k} 1_{\{G(\xi_i) \le x\}},$$
$$F_{k+1,n}(x) = \frac{1}{n-k} \sum_{i=k+1}^{n} 1_{\{G(\xi_i) \le x\}}.$$

For $\lambda \le \tau$, this yields the following representation:

$$\sum_{i=1}^{\lfloor n\lambda \rfloor} \sum_{j=\lfloor n\tau \rfloor+1}^{n} 1_{\{G(\xi_i) \le G(\xi_j)+\Delta\}} = (n - \lfloor n\tau \rfloor)\, \lfloor n\lambda \rfloor \int_{\mathbb{R}} F_{\lfloor n\lambda \rfloor}(x+\Delta) dF_{\lfloor n\tau \rfloor+1,n}(x).$$

Since $\frac{n-\lfloor n\tau \rfloor}{n} \longrightarrow 1-\tau$, it suffices to show that $\frac{\lfloor n\lambda \rfloor}{n} \int_{\mathbb{R}} F_{\lfloor n\lambda \rfloor}(x+\Delta) dF_{\lfloor n\tau \rfloor+1,n}(x)$ converges to $\lambda \int_{\mathbb{R}} F(x+\Delta) dF(x)$. For this purpose, we consider the inequality

$$\begin{aligned}
&\sup_{0\le\lambda\le\tau} \left| \frac{1}{n} \lfloor n\lambda \rfloor \int_{\mathbb{R}} F_{\lfloor n\lambda \rfloor}(x+\Delta) dF_{\lfloor n\tau \rfloor+1,n}(x) - \lambda \int_{\mathbb{R}} F(x+\Delta) dF(x) \right| \\
&\le \sup_{0\le\lambda\le\tau} \left| \frac{1}{n} \int_{\mathbb{R}} \lfloor n\lambda \rfloor F_{\lfloor n\lambda \rfloor}(x+\Delta) dF_{\lfloor n\tau \rfloor+1,n}(x) - \frac{\lfloor n\lambda \rfloor}{n} \int_{\mathbb{R}} F(x+\Delta) dF_{\lfloor n\tau \rfloor+1,n}(x) \right| \\
&\quad + \sup_{0\le\lambda\le\tau} \left| \frac{\lfloor n\lambda \rfloor}{n} \int_{\mathbb{R}} F(x+\Delta) dF_{\lfloor n\tau \rfloor+1,n}(x) - \lambda \int_{\mathbb{R}} F(x+\Delta) dF_{\lfloor n\tau \rfloor+1,n}(x) \right| \\
&\quad + \sup_{0\le\lambda\le\tau} \left| \lambda \int_{\mathbb{R}} F(x+\Delta) dF_{\lfloor n\tau \rfloor+1,n}(x) - \lambda \int_{\mathbb{R}} F(x+\Delta) dF(x) \right|.
\end{aligned} \tag{A2}$$

We will show that each of the three terms on its right-hand side converges to 0.

For the third summand, we obtain

$$\begin{aligned}
\sup_{0\le\lambda\le\tau} \left| \lambda \int_{\mathbb{R}} F(x+\Delta) dF_{\lfloor n\tau \rfloor+1,n}(x) - \lambda \int_{\mathbb{R}} F(x+\Delta) dF(x) \right| &= \tau \left| \int_{\mathbb{R}} \left(F_{\lfloor n\tau \rfloor+1,n}(x) - F(x)\right) dF(x+\Delta) \right| \\
&\le \tau \sup_{x\in\mathbb{R}} |F_{\lfloor n\tau \rfloor+1,n}(x) - F(x)|
\end{aligned}$$

as a consequence of integration by parts. Furthermore, we have

$$\sup_{x\in\mathbb{R}} |F_n(x) - F(x)| \longrightarrow 0 \text{ a.s.}$$

by an application of the Glivenko–Cantelli theorem (Krengel and Brunel, 1985) to the stationary and ergodic process $(G(\xi_i))_{i\ge1}$. So as to deduce an analogous result for $F_{\lfloor n\tau \rfloor+1,n}$, we rewrite

$$F_{\lfloor n\tau \rfloor+1,n}(x) = \frac{n}{n-\lfloor n\tau \rfloor} F_n(x) - \frac{\lfloor n\tau \rfloor}{n-\lfloor n\tau \rfloor} F_{\lfloor n\tau \rfloor}(x),$$

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and we may therefore conclude

$$|F_{\lfloor n\tau\rfloor+1,n}(x) - F(x)| \leq \left|\frac{n}{n-\lfloor n\tau\rfloor}\right| |F_n(x) - F(x)| + \left|\frac{\lfloor n\tau\rfloor}{n-\lfloor n\tau\rfloor}\right| |F_{\lfloor n\tau\rfloor}(x) - F(x)|.$$

Thus,

$$\sup_{x\in\mathbb{R}} |F_{\lfloor n\tau\rfloor+1,n}(x) - F(x)| \longrightarrow 0 \text{ a.s.,} \tag{A3}$$

which implies that the third term on the right-hand side of (A2) converges to 0 almost surely.

Regarding the second term on the right-hand side of (A2), we obtain

$$\begin{aligned}
&\sup_{0\leq\lambda\leq\tau} \left|\frac{\lfloor n\lambda\rfloor}{n}\int_{\mathbb{R}} F(x+\Delta)dF_{\lfloor n\tau\rfloor+1,n}(x) - \lambda\int_{\mathbb{R}} F(x+\Delta)dF_{\lfloor n\tau\rfloor+1,n}(x)\right| \\
&= \sup_{0\leq\lambda\leq\tau} \left|\frac{\lfloor n\lambda\rfloor}{n} - \lambda\right| \left|\int_{\mathbb{R}} F(x+\Delta)dF_{\lfloor n\tau\rfloor+1,n}(x)\right|.
\end{aligned}$$

The right-hand side of this equation converges to 0 since $\left|\int_{\mathbb{R}} F(x+\Delta)dF_{\lfloor n\tau\rfloor+1,n}(x)\right|$ is bounded by 1, and as $\sup_{0\leq\lambda\leq\tau}\left|\frac{\lfloor n\lambda\rfloor}{n} - \lambda\right| \longrightarrow 0$.

In order to show that the first term in (A2) converges to 0 as well, we consider the following inequality:

$$\begin{aligned}
&\sup_{0\leq\lambda\leq\tau} \left|\frac{1}{n}\int_{\mathbb{R}} \lfloor n\lambda\rfloor F_{\lfloor n\lambda\rfloor}(x+\Delta)dF_{\lfloor n\tau\rfloor+1,n}(x) - \frac{\lfloor n\lambda\rfloor}{n}\int_{\mathbb{R}} F(x+\Delta)dF_{\lfloor n\tau\rfloor+1,n}(x)\right| \\
&\leq \sup_{0\leq\lambda\leq\tau} \left|\frac{d_n}{n}\int_{\mathbb{R}} d_n^{-1}\lfloor n\lambda\rfloor\left(F_{\lfloor n\lambda\rfloor}(x+\Delta) - F(x+\Delta)\right) - J(x+\Delta)Z(\lambda)dF_{\lfloor n\tau\rfloor+1,n}(x)\right| \\
&\quad + \sup_{0\leq\lambda\leq\tau} \left|\frac{d_n}{n}Z(\lambda)\int_{\mathbb{R}} J(x+\Delta)dF_{\lfloor n\tau\rfloor+1,n}(x)\right|.
\end{aligned} \tag{A4}$$

In what follows, we will prove that both terms on the right-hand side of (A4) converge to 0. For this purpose, we make use of the empirical process non-central limit theorem of Dehling and Taqqu (1989), which states that

$$\left(d_n^{-1}\lfloor n\lambda\rfloor(F_{\lfloor n\lambda\rfloor}(x) - F(x))\right)_{x\in[-\infty,\infty],\lambda\in[0,1]} \xrightarrow{\mathcal{D}} J(x)Z(\lambda),$$

where '$\xrightarrow{\mathcal{D}}$' denotes convergence in distribution with respect to the σ-field generated by the open balls in $D([-\infty,\infty]\times[0,1])$, equipped with the supremum norm. Because of the Dudley–Wichura version of Skorohod's representation theorem (Shorack and Wellner, 1986, Theorem 2.3.4), we may assume without loss of generality that

$$\sup_{\lambda\in[0,1],x\in\mathbb{R}} \left|d_n^{-1}\lfloor n\lambda\rfloor\left(F_{\lfloor n\lambda\rfloor}(x) - F(x)\right) - J(x)Z(\lambda)\right| \longrightarrow 0$$

almost surely (Dehling *et al.*, 2013a). As a consequence, the first summand in (A4) converges to 0 since

$$\begin{aligned}
&\sup_{0\leq\lambda\leq\tau} \left|\frac{d_n}{n}\int_{\mathbb{R}} d_n^{-1}\lfloor n\lambda\rfloor\left(F_{\lfloor n\lambda\rfloor}(x+\Delta) - F(x+\Delta)\right) - J(x+\Delta)Z(\lambda)dF_{\lfloor n\tau\rfloor+1,n}(x)\right| \\
&\leq \frac{d_n}{n}\sup_{0\leq\lambda\leq\tau,x\in\mathbb{R}} \left|d_n^{-1}\lfloor n\lambda\rfloor\left(F_{\lfloor n\lambda\rfloor}(x+\Delta) - F(x+\Delta)\right) - J(x+\Delta)Z(\lambda)\right|
\end{aligned}$$

and as $\frac{d_n}{n}$ converges to 0 as well.

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For the second summand, we obtain the following inequality:

$$\sup_{0\le\lambda\le\tau}\left|\frac{d_n}{n}Z(\lambda)\int_{\mathbb{R}}J(x+\Delta)dF_{\lfloor n\tau\rfloor+1,n}(x)\right|\le\frac{d_n}{n}\sup_{0\le\lambda\le\tau}|Z(\lambda)|\left|\int_{\mathbb{R}}J(x+\Delta)dF_{\lfloor n\tau\rfloor+1,n}(x)\right|.$$

Note that

$$\begin{aligned}J(x)&=\int_{\mathbb{R}}1_{\{G(y)\le x\}}H_m(y)\varphi(y)dy\\&=-\int_{\mathbb{R}}1_{\{x\le G(y)\}}H_m(y)\varphi(y)dy,\end{aligned}$$

where φ denotes the standard normal density function, since

$$\int_{\mathbb{R}}H_m(y)\varphi(y)dy=0.$$

For this reason, we have

$$\begin{aligned}\int_{\mathbb{R}}J(x+\Delta)dF_{\lfloor n\tau\rfloor+1,n}(x)&=-\int_{\mathbb{R}}\int_{\mathbb{R}}1_{\{x+\Delta\le G(y)\}}H_m(y)\varphi(y)dydF_{\lfloor n\tau\rfloor+1,n}(x)\\&=-\int_{\mathbb{R}}\int_{\mathbb{R}}1_{\{x+\Delta\le G(y)\}}dF_{\lfloor n\tau\rfloor+1,n}(x)H_m(y)\varphi(y)dy\\&=-\int_{\mathbb{R}}F_{\lfloor n\tau\rfloor+1,n}(G(y)-\Delta)H_m(y)\varphi(y)dy.\end{aligned}$$

Analogous arguments yield

$$\int_{\mathbb{R}}J(x+\Delta)dF(x)=-\int_{\mathbb{R}}F(G(y)-\Delta)H_m(y)\varphi(y)dy.$$

Regarding the difference of these terms, we obtain

$$\begin{aligned}&\left|\int_{\mathbb{R}}J(x+\Delta)dF_{\lfloor n\tau\rfloor+1,n}(x)-\int_{\mathbb{R}}J(x+\Delta)dF(x)\right|\\&\le\sup_{y\in\mathbb{R}}|F(G(y)-\Delta)-F_{\lfloor n\tau\rfloor+1,n}(G(y)-\Delta)|\int_{\mathbb{R}}|H_m(y)|\varphi(y)dy,\end{aligned}$$

where $\int_{\mathbb{R}}|H_m(y)|\varphi(y)dy<\infty$ because of Hölder's inequality. As a result, $\int_{\mathbb{R}}J(x+\Delta)dF_{\lfloor n\tau\rfloor+1,n}(x)\xrightarrow{\mathcal{D}}\int_{\mathbb{R}}J(x+\Delta)dF(x)$ by (A3), so that, in the end, the second summand in (A4) converges to 0 almost surely as well. All in all, the third term on the right-hand side of (A2) converges to 0 almost surely as it is dominated by the sum of two expressions, which both converge to 0 with probability 1. This completes the proof of the first assertion in Lemma 1. □

Lemma 2. Suppose that $(\xi_i)_{i\ge1}$ is a stationary, long-range dependent Gaussian process with mean 0, variance 1 and long-range dependence parameter $0<D<\frac{1}{m}$, where m denotes the Hermite rank of the class of functions

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$1_{\{G(\xi_1)\le x\}} - F(x)$, $x \in \mathbb{R}$. Moreover, assume that $(G(\xi_i))_{i\ge 1}$ has a continuous distribution function F and that $G : \mathbb{R} \longrightarrow \mathbb{R}$ is a measurable function. Then

$$\frac{1}{n^2}\sum_{i=1}^{\lfloor n\tau\rfloor}\sum_{j=\lfloor n\tau\rfloor+1}^{\lfloor n\lambda\rfloor} 1_{\{G(\xi_i)\le G(\xi_j)+\Delta\}} \xrightarrow{P} \tau(\lambda-\tau)\int_{\mathbb{R}} F(x+\Delta)dF(x)$$

for fixed τ, uniformly in $\lambda \ge \tau$.

Proof of Lemma 2. An application of the triangle inequality yields

$$\begin{aligned}
&\sup_{\tau\le\lambda\le 1}\left|\frac{1}{n^2}\sum_{i=1}^{\lfloor n\tau\rfloor}\sum_{j=\lfloor n\tau\rfloor+1}^{\lfloor n\lambda\rfloor} 1_{\{G(\xi_i)\le G(\xi_j)+\Delta\}} - \tau(\lambda-\tau)\int_{\mathbb{R}} F(x+\Delta)dF(x)\right| \\
&\le \left|\frac{1}{n^2}\sum_{i=1}^{\lfloor n\tau\rfloor}\sum_{j=\lfloor n\tau\rfloor+1}^{n} 1_{\{G(\xi_i)\le G(\xi_j)+\Delta\}} - \tau(1-\tau)\int_{\mathbb{R}} F(x+\Delta)dF(x)\right| \\
&\quad + \sup_{\tau\le\lambda\le 1}\left|\frac{1}{n^2}\sum_{i=1}^{\lfloor n\tau\rfloor}\sum_{j=\lfloor n\lambda\rfloor+1}^{n} 1_{\{G(\xi_i)\le G(\xi_j)+\Delta\}} - \tau(1-\lambda)\int_{\mathbb{R}} F(x+\Delta)dF(x)\right|.
\end{aligned}$$

Lemma 2 then follows, because due to Lemma 1, the expressions on the right-hand side of the aforementioned inequality converge in probability to 0. □

Proof of Theorem 2. We have

$$\begin{aligned}
T_n(\tau_1,\tau_2) &= \sup_{k\in\{\lfloor n\tau_1\rfloor,\dots,\lfloor n\tau_2\rfloor\}} |G_n(k)| \\
&\ge \left|G_n(k^*)\right|,
\end{aligned}$$

where

$$G_n(k^*) = \frac{\frac{1}{n^2}\sum_{i=1}^{k^*}\sum_{j=k^*+1}^{n}\left(1_{\{X_i\le X_j\}} - \frac{1}{2}\right)}{\frac{1}{n^2}\left\{\frac{1}{n}\sum_{t=1}^{k^*} S_t^2(1,k^*) + \frac{1}{n}\sum_{t=k^*+1}^{n} S_t^2(k^*+1,n)\right\}^{\frac{1}{2}}}$$

and where $k^* = \lfloor n\tau\rfloor$ denotes the location of the change-point. Thus, it suffices to show that $|G_n(k^*)| \xrightarrow{P} \infty$. We will prove that the numerator of $G_n(k^*)$ converges to a positive constant, whereas the denominator tends to 0 in order to show divergence to ∞.

First, we turn to the denominator, which equals

$$\frac{1}{n^2}\left\{\int_0^{\tau} S_{\lfloor nr\rfloor}^2(1,k^*)dr + \int_{\tau}^{1} S_{\lfloor nr\rfloor}^2(k^*+1,n)dr\right\}^{\frac{1}{2}}.$$

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Note that

$$S_t(1,k^*) = \sum_{i=1}^{t}\sum_{j=k^*+1}^{n}\left(1_{\{G(\xi_i)\le G(\xi_j)\}} - \frac{1}{2}\right) - \sum_{i=1}^{t}\sum_{j=t+1}^{n}\left(1_{\{G(\xi_i)\le G(\xi_j)\}} - \frac{1}{2}\right)$$
$$- \sum_{i=1}^{t}\sum_{j=k^*+1}^{n} 1_{\{G(\xi_i)\le G(\xi_j)+\Delta\}} + \frac{t}{k^*}\sum_{i=1}^{k^*}\sum_{j=k^*+1}^{n} 1_{\{G(\xi_i)\le G(\xi_j)+\Delta\}}.$$

We have

$$\frac{1}{n^2}\sum_{i=1}^{\lfloor n\lambda\rfloor}\sum_{j=\lfloor n\lambda\rfloor+1}^{n}\left(1_{\{G(\xi_i)\le G(\xi_j)\}} - \frac{1}{2}\right) \xrightarrow{P} 0$$

uniformly in λ because

$$\frac{1}{nd_n}\sum_{i=1}^{\lfloor n\lambda\rfloor}\sum_{j=\lfloor n\lambda\rfloor+1}^{n}\left(1_{\{G(\xi_i)\le G(\xi_j)\}} - \frac{1}{2}\right) \xrightarrow{\mathcal{D}} (Z(\lambda) - \lambda Z(1))\int_{\mathbb{R}} J(x)dF(x)$$

uniformly in λ by Theorem 1.1 in Dehling *et al.* (2013a) and as $\frac{d_n}{n} \longrightarrow 0$. We may conclude from this and Lemma 1 that $\frac{1}{n^2}S_{\lfloor n\lambda\rfloor}(1,k^*) \xrightarrow{P} 0$ uniformly in $\lambda \le \tau$. The previous considerations yield

$$\int_0^\tau \left(\frac{1}{n^2}S_{\lfloor nr\rfloor}(1,k^*)\right)^2 dr \xrightarrow{P} 0$$

as $G : D[0,\tau] \longrightarrow \mathbb{R}$, $f \mapsto \int_0^\tau (f(s))^2\, ds$, is continuous with respect to the supremum norm on $D[0,\tau]$.

In analogy to the previous argumentation, it can be shown that

$$\int_\tau^1 \left(\frac{1}{n^2}S_{\lfloor nr\rfloor}(k^*+1,n)\right)^2 dr \xrightarrow{P} 0.$$

For this purpose, note that

$$S_t(k^*+1,n) = -\sum_{i=1}^{t}\sum_{j=t+1}^{n}\left(1_{\{G(\xi_i)\le G(\xi_j)\}} - \frac{1}{2}\right) + \sum_{i=1}^{k^*}\sum_{j=t+1}^{n}\left(1_{\{G(\xi_i)\le G(\xi_j)\}} - \frac{1}{2}\right)$$
$$+ \sum_{j=1}^{k^*}\sum_{i=k^*+1}^{t} 1_{\{G(\xi_j)\le G(\xi_i)+\Delta\}} - \frac{t-k^*}{n-k^*}\sum_{j=1}^{k^*}\sum_{i=k^*+1}^{n} 1_{\{G(\xi_j)\le G(\xi_i)+\Delta\}}.$$

Lemmas 1 and 2 then yield $\frac{1}{n^2}S_{\lfloor n\lambda\rfloor}(k^*+1,n) \xrightarrow{P} 0$ uniformly in $\lambda \ge \tau$.

On the basis of the previous considerations, we may conclude that the denominator of $G_n(k^*)$ converges in probability to 0. In order to prove consistency of the self-normalized Wilcoxon change-point test, it therefore remains to show that the numerator of $|G_n(k^*)|$ converges to a positive constant.

We have

$$\frac{1}{n^2}\left|\sum_{i=1}^{k^*}\sum_{j=k^*+1}^{n}\left(1_{\{X_i\le X_j\}} - \frac{1}{2}\right)\right| = \left|\frac{1}{n^2}\sum_{i=1}^{k^*}\sum_{j=k^*+1}^{n} 1_{\{G(\xi_i)\le G(\xi_j)+\Delta\}} - \frac{1}{2}\frac{k^*(n-k^*)}{n^2}\right|.$$

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Therefore,

$$\frac{1}{n^2}\left|\sum_{i=1}^{k^*}\sum_{j=k^*+1}^{n}\left(1_{\{X_i\le X_j\}}-\frac{1}{2}\right)\right| \xrightarrow{P} \tau(1-\tau)\int_{\mathbb{R}}|F(x+\Delta)-F(x)|\,dF(x) \tag{A5}$$

by Lemma 1 and since $\frac{1}{2}\frac{k^*(n-k^*)}{n^2} \longrightarrow \frac{\tau(1-\tau)}{2}$. As the limit in (A5) does not vanish, $|G_n(k^*)|$ diverges to ∞, and we thus have proved Theorem 2. □

Proof of Theorem 3. Note that because of the corresponding sample path properties of the stochastic process Z, the sample paths of

$$W^*_{m,\tau}(\lambda) = (Z(\lambda)-\lambda Z(1))\int_{\mathbb{R}} J(x)dF(x) + c\delta_\tau(\lambda)\int_{\mathbb{R}} f^2(x)dx,\ 0\le\lambda\le 1,$$

are almost surely continuous and nowhere differentiable. The same argument as in the proof of Theorem 1 shows that $T_n(\tau_1,\tau_2)$ converges in distribution to

$$\sup_{\lambda\in[\tau_1,\tau_2]}\frac{\left|W^*_{m,\tau}(\lambda)\right|}{\left\{\int_0^\lambda\left(W^*_{m,\tau}(r)-\frac{r}{\lambda}W^*_{m,\tau}(\lambda)\right)^2dr+\int_\lambda^1\left(W^*_{m,\tau}(r)-\frac{1-r}{1-\lambda}W^*_{m,\tau}(\lambda)\right)^2dr\right\}^{\frac{1}{2}}},$$

which corresponds to the expression in equation (6). □

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